MAPS INTO A BOUNDED BELOW COMPLEX OF FILTERED INJECTIVES UP TO FILTERED QUASI-ISOMORPHISM

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Editorial note. Counted span is the author's own proof environment, source lines 8482-8523, following the statement at 8457-8480.

Editorial note. No omissions: part (1) displays the exact Hom sequence in an xymatrix; part (2) uses (1) twice, once for surjectivity and once for injectivity.

Allowed standard prerequisites (editorial).

- Cone triangle in the homotopy category. lemma-the-same-up-to-isomorphisms (carded L26170, cited around line 8490). Provides (K, L, C(alpha), alpha, i, -p) as a distinguished triangle in K(Fil^f(A)).
- Filtered acyclicity of cones and its Hom consequence. lemma-filtered-acyclic (cited around 8492) and lemma-filtered-acyclic-is-zero (cited around 8505), both same chapter. Identifies C(alpha) as filtered acyclic and kills the two outer Hom groups.
- Representable functor is homological. lemma-representable-homological (same chapter, cited around line 8503). Gives the four-term exact Hom sequence attached to the cone triangle.

Source reference. Original lemma 05S1: sources/stacks-derived/pinned-original-source.tex, source lines 4509–4509. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05TX: sources/stacks-derived/pinned-original-source.tex, source lines 8414–8414. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0149: sources/stacks-derived/pinned-original-source.tex, source lines 603–603. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 014L: sources/stacks-derived/pinned-original-source.tex, source lines 3213–3213. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $I^\bullet \in K(Fil^f(\mathcal{A}))$ be a bounded below complex consisting of filtered injective objects.*

(1) *Let $\alpha : K^\bullet \to L^\bullet$ in $K(Fil^f(\mathcal{A}))$ be a filtered quasi-isomorphism. Then the map*
$$\mathrm{Hom}_{K(Fil^f(\mathcal{A}))}(L^\bullet, I^\bullet) \to \mathrm{Hom}_{K(Fil^f(\mathcal{A}))}(K^\bullet, I^\bullet)$$
is bijective.

(2) *Let $L^\bullet \in K(Fil^f(\mathcal{A}))$. Then*
$$\mathrm{Hom}_{K(Fil^f(\mathcal{A}))}(L^\bullet, I^\bullet) = \mathrm{Hom}_{DF(\mathcal{A})}(L^\bullet, I^\bullet).$$

Proof. Proof of (1). Note that
$$(K^\bullet, L^\bullet, C(\alpha)^\bullet, \alpha, i, -p)$$
is a distinguished triangle in $K(\mathrm{Fil}^f(\mathcal{A}))$ (Lemma 014L) and $C(\alpha)^\bullet$ is a filtered acyclic complex (Lemma 05S1). Then
$$\mathrm{Hom}_{K(\mathrm{Fil}^f(\mathcal{A}))}(C(\alpha)^\bullet, I^\bullet) \longrightarrow \mathrm{Hom}_{K(\mathrm{Fil}^f(\mathcal{A}))}(L^\bullet, I^\bullet) \longrightarrow \mathrm{Hom}_{K(\mathrm{Fil}^f(\mathcal{A}))}(K^\bullet, I^\bullet)$$
$$\mathrm{Hom}_{K(\mathrm{Fil}^f(\mathcal{A}))}(C(\alpha)^\bullet[-1], I^\bullet)$$
is an exact sequence of abelian groups, see Lemma 0149. At this point Lemma 05TX guarantees that the outer two groups are zero and hence $\mathrm{Hom}_{K(\mathcal{A})}(L^\bullet, I^\bullet) = \mathrm{Hom}_{K(\mathcal{A})}(K^\bullet, I^\bullet)$.

Proof of (2). Let a be an element of the right hand side. We may represent $a = \gamma\alpha^{-1}$ where $\alpha : K^\bullet \to L^\bullet$ is a filtered quasi-isomorphism and $\gamma : K^\bullet \to I^\bullet$ is a map of complexes. By part (1) we can find a morphism $\beta : L^\bullet \to I^\bullet$ such that $\beta \circ \alpha$ is homotopic to γ. This proves that the map is surjective. Let b be an element of the left hand side which maps to zero in the right hand side. Then b is the homotopy class of a morphism $\beta : L^\bullet \to I^\bullet$ such that there exists a filtered quasi-isomorphism $\alpha : K^\bullet \to L^\bullet$ with $\beta \circ \alpha$ homotopic to zero. Then part (1) shows that β is homotopic to zero also, i.e., $b = 0$. □